**Code No: 133BM** **R16**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech II Year I Semester Examinations, August/September - 2022**
**OBJECT ORIENTED PROGRAMMING THROUGH JAVA**
**(Common to CSE, IT)**

**Time: 3 Hours** **Max.Marks:75**

**Answer any five questions**
**All questions carry equal marks**
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1.a) What are object-oriented paradigms? Explain in detail.
b) Explain about various data types used in Java with examples. [8+7]

2.a) Discuss about various types of polymorphism.
b) What are Java buzzwords? Explain in detail. [8+7]

3.a) Explain about importing Java packages with suitable Java program.
b) Write a program to improve the efficiency of writing data into a file using Buffered Output Stream. [8+7]

4.a) Discuss about input and output streams in Java with examples.
b) Explain about implementation of interfaces in Java. [8+7]

5.a) What is an exception? Explain about exceptional handling in Java.
b) Explain about creation of threads in Java. [8+7]

6.a) Discuss about creating user defined exceptions in detail.
b) Differentiate between thread-based multitasking and process-based multitasking. [8+7]

7.a) Define HashMap. Write a program that shows the use of HashMap class.
b) Explain about Calendar class and DateFormat class methods in detail. [8+7]

8. Explain the following:
a) MVC architecture
b) Anonymous inner classes
c) Security issues in applets. [5+5+5]

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